

AuraCampus × Bootcoding Pvt. Ltd.

Tense AI Hackathon Presentation

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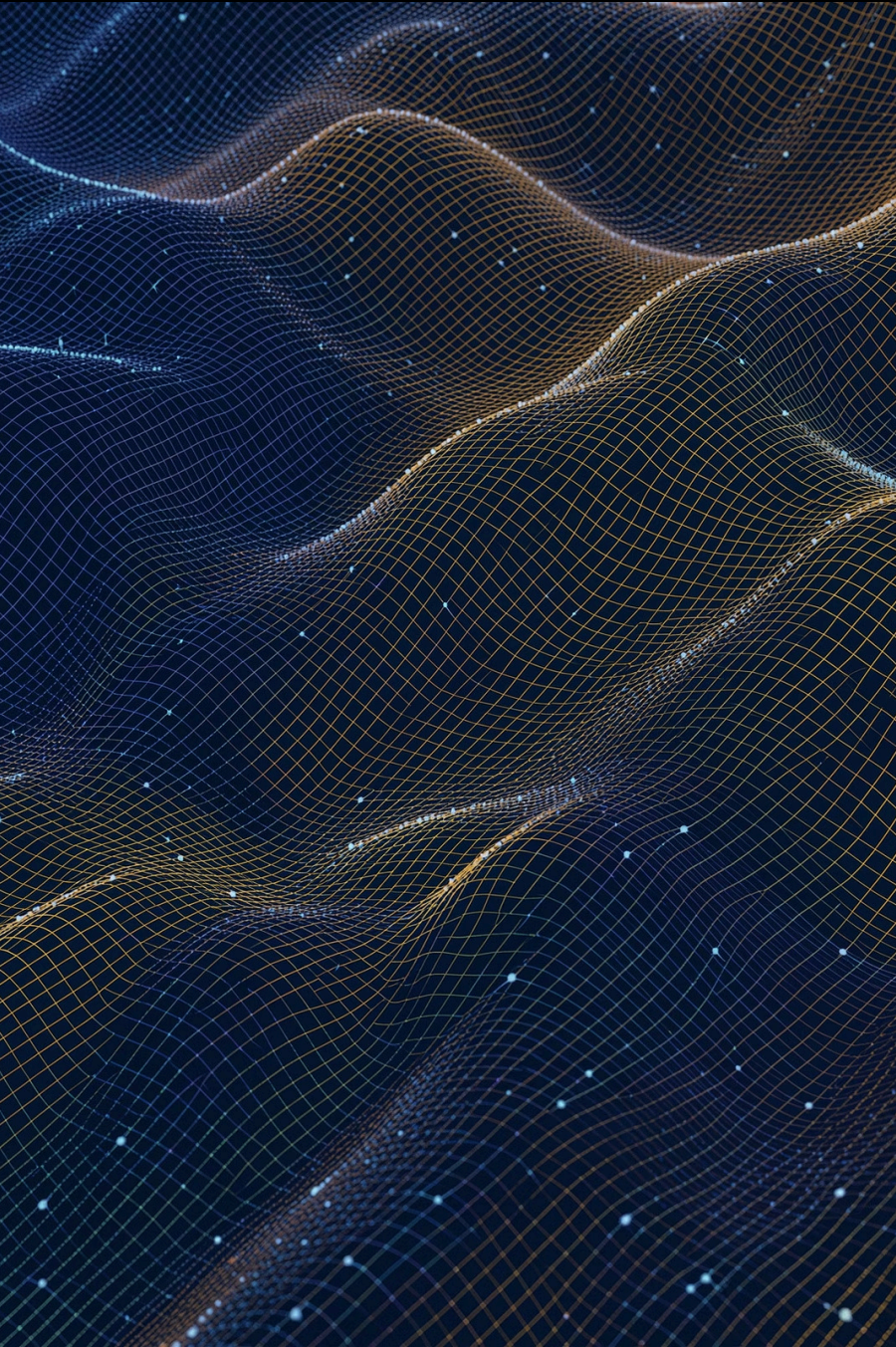
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PROJECT VISION

AuraCampus

Anonymous.

Intelligent. Empathetic.

A **privacy-first sentiment intelligence platform** that transforms anonymous campus check-ins into actionable wellness insights through dynamic data visualisation, context-aware AI, and non-alarmist intervention pathways.



Privacy-First

Zero-knowledge anonymity at every layer



Intelligent

Context-aware AI with explainable outputs



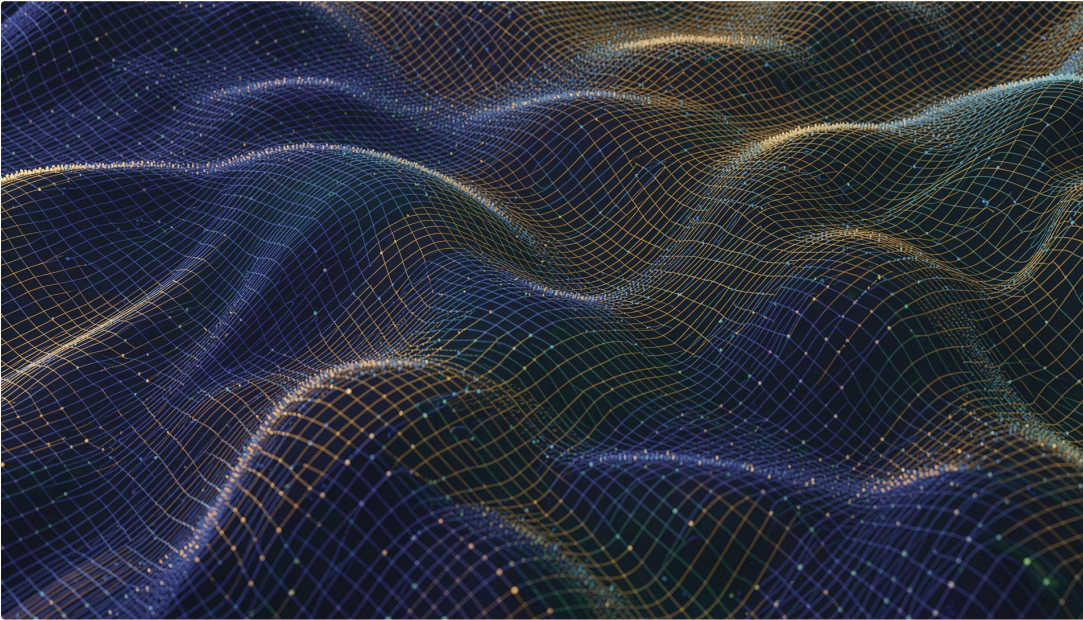
Empathetic

Non-alarmist, human-centred interventions

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Campus Mental Health: A Growing Crisis



The Challenge

Develop a solution that **monitors sentiment across campus communities** — through anonymous check-ins, forums, or feedback — and flags distress patterns for timely intervention.

⚠ Students in distress often go undetected until it is too late. Traditional systems lack the sensitivity, privacy, and speed required for early intervention.

- Anonymous check-ins remain underutilised due to trust gaps
- Counsellors lack real-time, aggregated distress signals
- Existing tools fail to distinguish noise from genuine crisis
- Privacy concerns deter honest self-reporting

Built for Scale, Speed & Safety



Frontend

- React Native (Expo) for mobile
- Next.js + Framer Motion for web
- Lottie animations + Three.js 3D visualisations



Backend

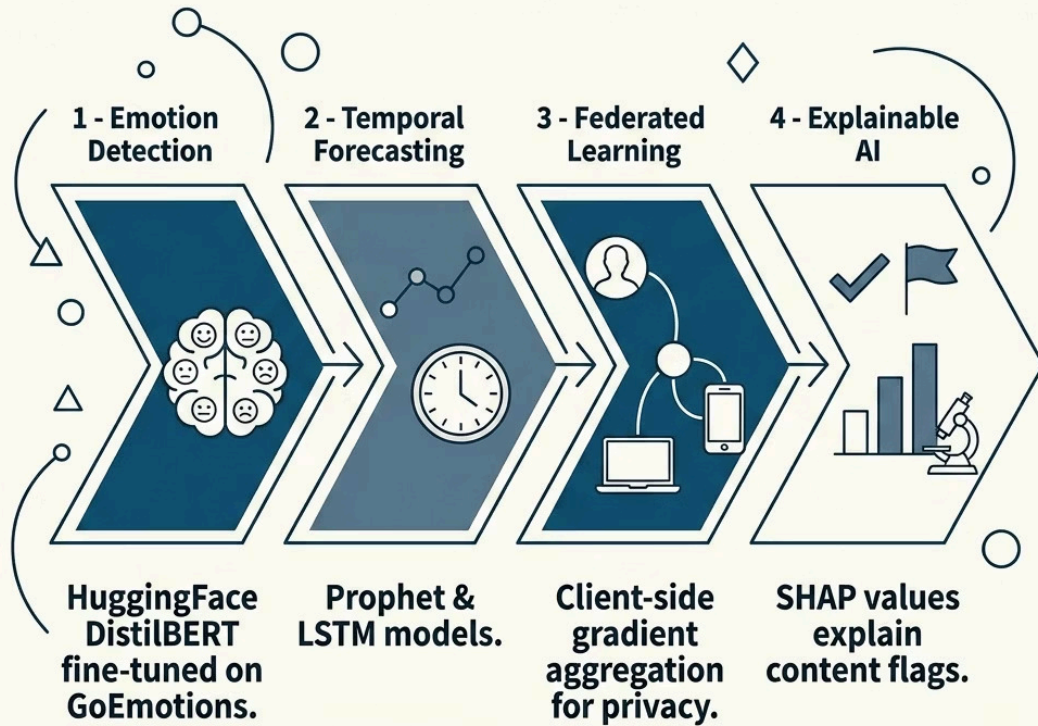
- FastAPI (Python) for high-performance APIs
- WebSocket streams for real-time updates
- Redis cache for low-latency responses



Database

- PostgreSQL for structured metadata
- TimescaleDB for time-series sentiment data

Intelligence at Every Layer



Why This Pipeline?

Each stage is purpose-built to balance **accuracy, privacy, and transparency** — the three pillars that make AuraCampus trustworthy at scale.

DistilBERT + QoEmotions

Fine-tuned for nuanced campus emotional vocabulary

Prophet / LSTM

Forecasts distress trends before they peak

Federated Learning

Model trains on-device — raw data never leaves the user

SHAP Explainability

Counsellors see *why* a flag was

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Enterprise-Grade Privacy by Design

Infrastructure

AWS ECS

Containerised, auto-scaling microservices

CloudFront

Global CDN for low-latency delivery

Vercel Edge

Serverless functions at the network edge

Security Stack

→ OAuth 2.0

Industry-standard authorisation framework

→ Zero-Knowledge Anonymity Layer

Identity is mathematically provably hidden

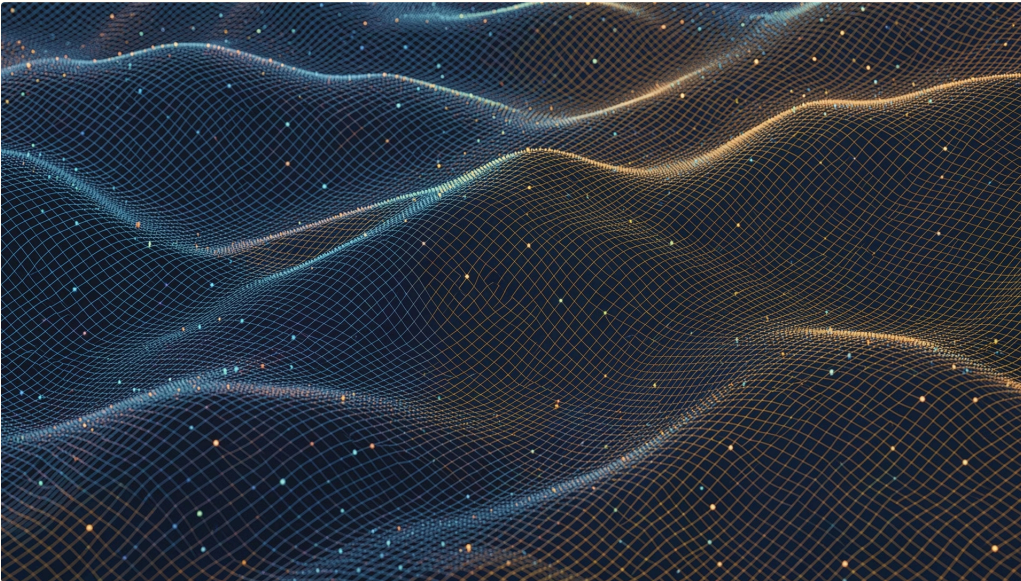
→ Differential Privacy

Statistical noise injection prevents re-identification

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Existing Solutions & Their Gaps



What Already Exists

Several initiatives promote mental health on campuses through digital tools and anonymous platforms. Apps and web portals enable students to complete regular check-ins, share experiences on forums, or provide feedback on stress and wellbeing.

Sentiment analysis and NLP techniques have been applied to detect anxiety, depression, or distress patterns in student submissions. Some platforms integrate automated alerts for counsellors when critical signals are identified, while others combine peer support with professional guidance.

⊗ Despite their promise, these solutions consistently face three critical challenges: **privacy**, **engagement**, and **accurate early detection**.

Privacy Gaps

Students distrust platforms that cannot guarantee true anonymity



Low Engagement

Passive tools fail to sustain consistent student participation



False Positives

Context-blind models misread exam stress as crisis signals

What Makes AuraCampus Different



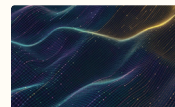
Mood Constellation Visualisation

Communities appear as star clusters. **Brightness = wellness volume**, pulse frequency = distress rate. Spatial metaphors that judges and users instantly understand.



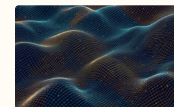
Context-Aware Baselines

AI adjusts sentiment thresholds using **academic calendar, local weather, and campus events** — preventing false positives during midterms or holidays.



Peer Resonance Network

Opt-in anonymous matching with users who shared similar wellness journeys. Focused on **lived experience**, not demographics.



Wellness Echo System

When a cluster improves, a visual **"ripple" flows across the map**, reinforcing positive community momentum and celebrating collective progress.

Tiered Intervention Matrix

AuraCampus never over-escalates. The AI suggests three calibrated levels of response:



Tier 1 — Resource Drop

Sleep hygiene guides, exam stress toolkits, and self-help content delivered contextually



Tier 2 — Counsellor Connect

Counsellor availability heatmap + opt-in anonymous chat for guided support



Tier 3 — Crisis Protocol

Direct hotline bypass + campus safety link for immediate, life-critical intervention

